LABSHEET-01

```
<!DOCTYPE html>
<html>
<head>
<title>Resume & Product Catalog</title>
<style>
*{box-sizing:border-box}
body{margin:0;font:16px Arial;line-height:1.5;background:#f4f4f4;color:#222}
nav{background:#222}
nav ul{display:flex;margin:0;padding:0;list-style:none}
nav li{position:relative}
nav a{display:block;padding:14px 20px;color:white;text-decoration:none}
nav a:hover{background:#555}
nav ul ul{display:none;position:absolute;background:#333;padding:0}
nav li:hover>ul{display:block}
nav ul ul ul{left:100%;top:0}

header{background:#1f2937;color:white;text-align:center;padding:30px}
section{max-width:900px;margin:20px auto;padding:25px;background:white;border-radius:8px}
h2{color:#1f2937;border-bottom:2px solid #ddd;padding-bottom:6px}
.catalog{display:grid;grid-template-columns:repeat(3,1fr);gap:20px}
.product{padding:18px;border:1px solid #ccc;border-radius:8px;background:#fafafa}
dt{font-weight:bold;font-size:18px;margin-top:8px}
dd{margin:0 0 10px;color:#555}
.price{color:green;font-weight:bold}
footer{text-align:center;background:#222;color:white;padding:20px}

@media(max-width:700px){
nav ul{flex-direction:column}
nav ul ul{position:static}
.catalog{grid-template-columns:1fr}
}
</style>
</head>

<body>

<nav>
<ul>
<li><a href="#">Home</a></li>
<li><a href="#">About</a>
<ul>
<li><a href="#">Our Team</a></li>
<li><a href="#">History</a></li>
</ul>
</li>
<li><a href="#">Services</a>
```

```
<ul>
<li><a href="#">Web Development</a></li>
<li><a href="#">AI</a>
<ul>
<li><a href="#">Machine Learning</a></li>
<li><a href="#">Deep Learning</a></li>
</ul>
</li>
</ul>
</li>
<li><a href="#">Resources</a></li>
<li><a href="#">Contact</a></li>
</ul>
</nav>

<header>
<h1>Shiv Bhatia</h1>
<p>B.Tech Computer Science Student</p>
<p>Email: shivbhatia38@gmail.com | India</p>
</header>

<main>

<section>
<h2>About Me</h2>
<p>Computer Science student interested in AI, ML, Web Development and Data Structures.</p>
</section>

<section>
<h2>Education</h2>
<h3>B.Tech in Computer Science</h3>
<p>COER University | 2024 - 2028</p>
</section>

<section>
<h2>Technical Skills</h2>
<ul>
<li>C++</li><li>Python</li><li>HTML & CSS</li>
<li>JavaScript</li><li>Machine Learning</li><li>MongoDB</li>
</ul>
</section>
<section>
<h2>Projects</h2>
<h3>AI Nutrition Planner</h3>
<p>ML application for personalized nutrition recommendations.</p>
<h3>Automatic License Plate Recognition</h3>
<p>AI system for detecting and recognizing vehicle plates.</p>
</section>
```

```
<section>
<h2>Computer Product Catalog</h2>
<div class="catalog">

<article class="product">
<dl>
<dt>Laptop</dt><dd>Ryzen 5, 16GB RAM, 512GB SSD</dd>
<dt>Price</dt><dd class="price">₹55,000</dd>
</dl>
</article>

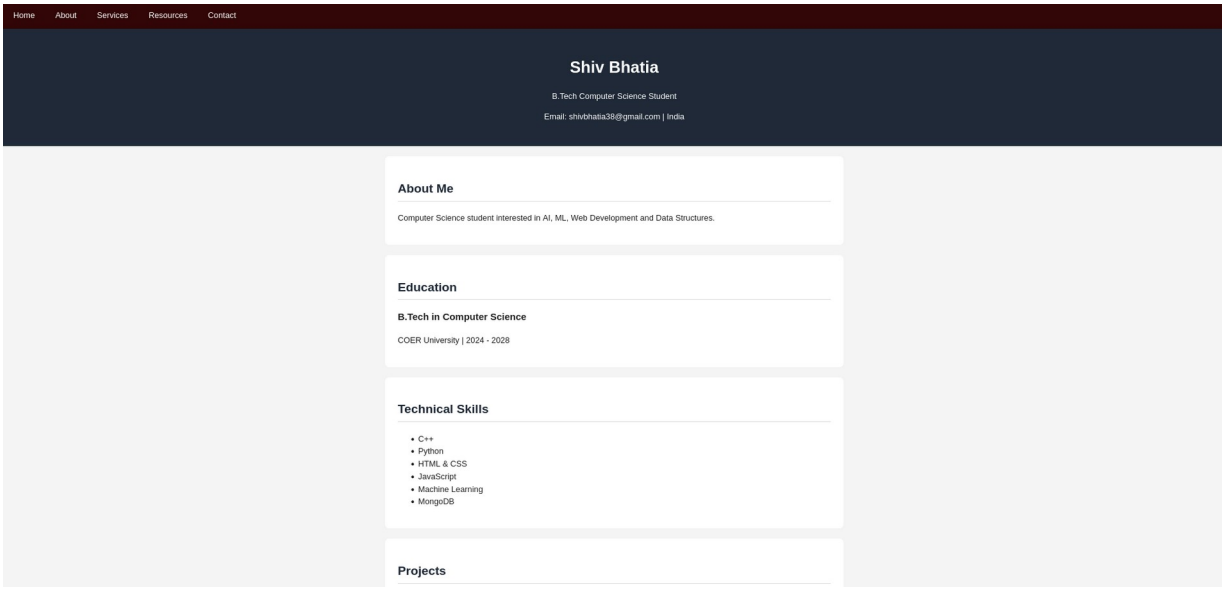
<article class="product">
<dl>
<dt>Mechanical Keyboard</dt><dd>Blue Switch, USB, RGB</dd>
<dt>Price</dt><dd class="price">₹2,500</dd>
</dl>
</article>

<article class="product">
<dl>
<dt>Wireless Mouse</dt><dd>Bluetooth, 1600 DPI, Rechargeable</dd>
<dt>Price</dt><dd class="price">₹1,200</dd>
</dl>
</article>

</div>
</section>

</main>

<footer>
<p>© 2026 Shiv Bhatia | Contact Me</p>
</footer>
</body>
</html>
```



LABSHEET-02

```
<!DOCTYPE html>
<html>
<head>
<title>Interactive Website</title>
<style>
*{box-sizing:border-box}
body{
 margin:0;font-family:Arial;text-align:center;
 background:#f4f4f4;color:#222;
 transition:.3s
}
.dark{background:#121212;color:white}
nav{
 padding:20px;display:flex;
 justify-content:space-between
}
button{
 padding:10px 15px;border:0;border-radius:6px;
 cursor:pointer
}
.hero{padding:80px 20px}
.primary{background:#2563eb;color:white}

.gallery{padding:20px}
.grid{
 display:flex;flex-wrap:wrap;
 justify-content:center;gap:20px
}
.card{
 width:280px;overflow:hidden;
 border-radius:10px;
 transition:.3s
}
.card:hover{transform:scale(1.05)}
.card img{width:100%;height:200px;object-fit:cover}

.todo{max-width:600px;margin:40px auto;padding:20px}
input{padding:12px;width:70%}
li{
 list-style:none;margin:10px;padding:10px;
 background:#ddd;display:flex;
 justify-content:space-between
}
.done{text-decoration:line-through}
#error{color:red}
</style>
```

```
</head>

<body>

<nav>
<h2>My Website</h2>
<button onclick="dark()">🌙 Dark Mode</button>
</nav>

<section class="hero">
<h1>Welcome to My Interactive Website</h1>
<p>HTML, CSS and JavaScript Demo</p>
<button class="primary" onclick="msg()">Click Me</button>
<p id="message"></p>
</section>

<section class="gallery">
<h2>Responsive Photo Gallery</h2>
<div class="grid">
<img class="card" src="https://picsum.photos/id/1015/400/300">
<img class="card" src="https://picsum.photos/id/1016/400/300">
<img class="card" src="https://picsum.photos/id/1025/400/300">
<img class="card" src="https://picsum.photos/id/1035/400/300">
<img class="card" src="https://picsum.photos/id/1043/400/300">
<img class="card" src="https://picsum.photos/id/1069/400/300">
</div>
</section>

<section class="todo">
<h2>Todo Application</h2>
<input id="task" placeholder="Enter task">
<button class="primary" onclick="add()">Add</button>
<p id="error"></p>
<ul id="list"></ul>
</section>

<script>
function dark(){
 document.body.classList.toggle("dark");
}

function msg(){
 document.getElementById("message").innerText=
 "JavaScript is working successfully!";
}

function add(){
 let t=document.getElementById("task");
 let e=document.getElementById("error");
```

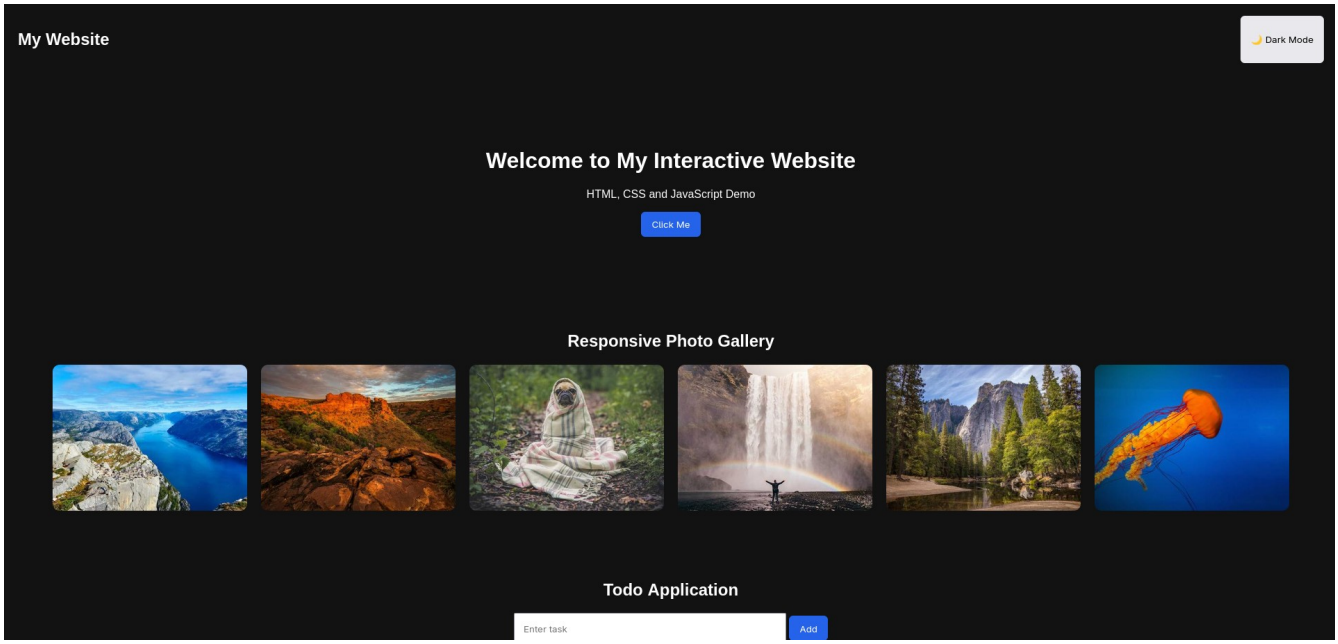
```
 if(!t.value.trim()){
  e.innerText="Please enter a task";
  return;
 }

 e.innerText="";
 let li=document.createElement("li");
 li.innerHTML=`<span>${t.value}</span>
 <span>
 <button onclick="this.parentElement.parentElement
 .classList.toggle('done')">Done</button>
 <button onclick="this.parentElement.parentElement.remove()">
 Delete</button>
 </span>`;

 document.getElementById("list").appendChild(li);
 t.value="";
}

document.getElementById("task").onkeydown=e=>{
 if(e.key==="Enter") add();
};
</script>

</body>
</html>
```



Lab 4 — React Routing and State Management

Aim: Build a multi-page single-page application (SPA).
Tasks:
Create a 3-page app (Home, Products, Product Details) using react-router-dom.
Pass data between routes using route params and/or Context API.
Implement a global state (e.g., cart or favorites) using Context API or Redux Toolkit (either allowed).
Add a loading state and a 404 Not Found page.

Deliverables: Deployed link or local demo video, source code.

CODE:-

```
import {useState,useEffect} from "react";

export default function App(){
  const [tasks,setTasks]=useState(
    JSON.parse(localStorage.getItem("tasks"))||[]
  );
  const [task,setTask]=useState("");

  useEffect(()=>{
    localStorage.setItem("tasks",JSON.stringify(tasks));
  },[tasks]);

  function addTask(e){
    e.preventDefault();
    if(task.trim()){
      setTasks([...tasks,{id:Date.now(),text:task}]);
      setTask("");
    }
  }

  return(
    <div style={{
      maxWidth:"500px",margin:"60px auto",padding:"20px",
      fontFamily:"Arial",textAlign:"center",
      background:"#f4f4f4",borderRadius:"10px"
    }}>
      <h1>Todo App</h1>

      <form onSubmit={addTask}>
        <input
          value={task}
          onChange={e=>setTask(e.target.value)}
          placeholder="Enter task"
          style={{padding:"10px"}}
        />
        <button>Add</button>
      </form>
```

```
      {tasks.map(t=>(
        <div key={t.id} style={{
          background:"white",padding:"10px",margin:"10px",
          display:"flex",justifyContent:"space-between"
        }}>
          {t.text}
          <button onClick={()=>
            setTasks(tasks.filter(x=>x.id!==t.id))
          }>Delete</button>
        </div>
      ))}
    </div>
  );
}
```

